

Telco Customer Churn Data Cleaning Report

A concise, evidence-based cleaning audit for a 7,043-customer telecommunications churn dataset. The report distinguishes genuine data-quality issues from validation checks that found no defect.

7,043	21	2	High
Customer records	Dataset fields	Genuine issues	Readiness

Executive summary

The dataset is structurally strong and suitable for exploratory analysis after a focused correction to **TotalCharges**. The column contains 11 whitespace-only entries and is consequently stored as text rather than a numeric monetary field. The affected records all have zero tenure, so a zero cumulative-charge value is business-plausible once the conversion is documented.

No duplicate records or customer identifiers were found. Service categories are internally coherent, no negative tenure or charge values were found, and IQR screening identified no statistical outliers in tenure, MonthlyCharges, or TotalCharges. These are validation findings, not additional problems.

Scope and decision rule

This report uses only findings established during the dataset exploration. A condition is listed as a data cleaning problem only where it requires a correction or a documented handling decision. Plausible business variation and checks with no defect are retained as validation evidence.

Recommended disposition

Standardize missing-value representation, convert TotalCharges to numeric, set the 11 zero-tenure blank totals to 0 only with the stated business rationale, then complete final validation and proceed to EDA.

Data Cleaning Problem #1: Hidden blank values in TotalCharges

REASONING	TotalCharges contains 11 whitespace-only values (0.16% of customer records). Standard missing-value scans can miss these entries because they are stored as text rather than nulls. Each affected record has tenure = 0, which supports the interpretation that no cumulative bill has yet been recorded. The rows also have valid current monthly charges; this is not contradictory because a plan can exist before the first billing cycle closes.
SOLUTION	Replace whitespace-only entries with a standard missing representation, convert the column to numeric, and then set the missing total to 0 for the zero-tenure records. Document the rule as a business-logic treatment of no accumulated billing history - not a general imputation method. Retain the original source file and record the 11-row change in the cleaning log.

Evidence supporting the disposition

Measure	Observed finding	Interpretation
Affected records	11 of 7,043	Small, fully identified subset
Tenure pattern	11 of 11 have tenure = 0	Consistent with no accumulated charges
Contract context	One year or Two year only	Descriptive context; not the causal explanation

Data Cleaning Problem #2: TotalCharges stored as text instead of numeric

REASONING	TotalCharges represents a monetary amount but is stored as an object/text field. The direct cause is the embedded whitespace-only values. Leaving the field as text can prevent or distort numeric summaries, distributions, correlations, aggregation, and outlier screening. It also creates an avoidable risk of lexicographic sorting, where values may be ordered as text instead of amounts.
SOLUTION	After standardizing the 11 blank entries, convert TotalCharges using a safe numeric conversion. Confirm that conversion errors are limited to the known blank records before applying the zero-tenure treatment. Use a numeric decimal-compatible type for analysis, then rerun descriptive statistics and missing-value checks as final verification.

Validation findings: retained values, no additional cleaning action

The following results support analysis readiness but are not listed as defects because they required no correction.

Validation area	Finding	Conclusion
Duplicates	0 duplicate rows; 0 duplicate customer IDs	No deletion or deduplication required
Text formatting and categories	No widespread extra-space or category-format issue; service labels align with service status	Preserve valid labels such as 'No phone service' and 'No internet service'
Impossible numeric values	0 records with negative tenure or charges	No remediation required
IQR outlier screen	0 outliers in tenure, MonthlyCharges, and TotalCharges	Retain observed values; no statistical outlier removal
Charge relationship check	59 records exceeded a 20% comparison threshold against tenure x current MonthlyCharges	Plausible historical billing or plan changes; do not alter or delete

Final cleaning checklist

1. Create and preserve a working copy. 2. Convert whitespace-only TotalCharges values to missing. 3. Convert TotalCharges to numeric. 4. Set only the 11 zero-tenure missing totals to 0 and log the rationale. 5. Recheck missing values, duplicate counts, data types, and numeric summaries. 6. Begin EDA without deleting valid categories, the 59 billing-deviation records, or any values based solely on an IQR screen.